PGPM 2026–27 | Session 06

Statistical Methods in Decision Making

Case Study Review & Critical Analysis

Nandan Mart: A Portfolio of Six Hypothesis Tests

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1. Verification of Sir' Solution

I went through every number in the case study — recomputing all six t-statistics from the raw dataset values. The short answer is: **sir's solution is arithmetically correct across all six hypotheses.** Every t-value, SE, and p-value checked out. That said, there maybe a few places where the solution is technically right but could be more transparent about what it is actually telling us.

1.1 Recomputed Summary Statistics (from Dataset)

Hyp.	Test Type	$\bar{x}$ or $\bar{d}$	s or s_a	SE	t (computed)	Faculty t	p-value	Verdict
H1	One-sample	32.20	1.740	0.449	**4.90**	4.90 ✓	0.0001	**Reject H_0**
H2	One-sample	4.007	0.171	0.044	**0.15**	0.15 ✓	0.882	**Fail to Reject**
H3	Two independents	182.13 vs 163.27	5.69 / 5.06	1.967	**9.59**	9.59 ✓	< 0.0001	**Reject H_0**
H4	Two independents	3.233 vs 3.127	0.229 / 0.231	0.084	**1.27**	1.27 ✓	0.107	**Fail to Reject**
H5	Paired	$\bar{d}$ = 21.07	3.218	0.831	**25.36**	25.36 ✓	< 0.0001	**Reject H_0**
H6	Paired	$\bar{d}$ = −0.467	0.915	0.236	**−1.97**	−1.97 ✓	0.034 (left)	**Reject H_0***

** H6 clears the one-tailed bar (p = 0.034) but not the two-tailed bar (p = 0.068). The one-tailed verdict is valid only because the direction was fixed in advance from the vendor's claim.*

1.3 Points for Refinement

H3 pooled variance assumption: The solution pools the variances (s_3 = 5.69, s_4 = 5.06) without explicitly checking whether they are comparable. They are — the ratio is about 1.26, well within the rule-of-thumb range — but the check is missing from the write-up.

H1 two-tailed omission: The case frames H1 as right-tailed, which is correct. But the solution does not show what the two-tailed result would look like. Given that H6 dedicates a whole section to the one-tail vs. two-tail comparison, applying the same stress-test to H1 would have made the pedagogy consistent.

H4 practical interpretation: The solution correctly fails to reject H4 but describes the 3.23% vs 3.13% gap as "within noise" without quantifying what a meaningful gap would look like. Given the 0.107 p-value — not 0.5, not 0.8 — this is a borderline miss. The correct framing is: fail to reject with n = 15, but the observed gap is not trivial; a properly powered study (roughly n = 50+) should be run before writing the website off.

2. Are All Six Tests Actually Needed?

The honest answer is: no, not all six are equally necessary for the business decision Devika faces. The question is which ones drive real money, and which ones are more about intellectual completeness than operational urgency.

2.1 The Three Tests That Actually Matter Most

H1 — Delivery time (most operationally urgent): This is the fire. A 2.2-minute average breach of a stated 30-minute promise, confirmed with a t of 4.90 and a p-value of 0.0001, is not a rounding error. It is a broken commitment to customers. Every other decision in this portfolio is downstream of whether the company can keep its core delivery promise.

H5 — Loyalty program (highest ROI signal): A t-statistic of 25.36 is so far past the cutoff that this is essentially a certainty within the constraints of n = 15. More importantly, the effect is consistent: every single difference in the paired data is positive, which means the basket increase was not driven by one or two outlier stores. This is the strongest result in the dataset, and the case for chain-wide rollout is as close to bulletproof as a 15-observation study can produce.

H6 — Route software (most nuanced): This test earns its place not because the result is strong, but because the decision is expensive. Renewing a software license on the basis of a 28-second improvement is hard to justify. The test confirms the software works in the technical sense, but the business question — is it worth renewing? — has to incorporate the cost of the license, not just the direction of the effect.

2.2 Tests That Could Be Deprioritized

H2 — CSAT: A p-value of 0.882 is about as unambiguous as statistical results get. The "slipping rating" narrative appears to be pure anecdote. This test is worth doing exactly once to kill the conversation — and the result does that effectively. But it does not require follow-up.

H3 — Urban vs. suburban: The gap is real (t = 9.59), but the test is measuring sales, not profit. Without margins, occupancy costs, and catchment data, the result cannot support a store closure decision. The test is useful for scoping a deeper study; it is not useful as a standalone basis for action.

H4 — App vs. web conversion: The most borderline result in the set. The app leads by 0.11 percentage points, the p-value is 0.107, and with n = 15 the test simply does not have the power to resolve a gap this small. This is not a test to drop — it is a test that says "run again with more data before you shift the marketing budget."

2.3 Verdict on Necessity

For an immediate operational decision: H1, H5, and H6 are the essential tests. H2 is useful for closing down a false narrative. H3 and H4 are better understood as scoping exercises that justify follow-on studies rather than standalone answers. All six together form a coherent portfolio — but the resource implications differ significantly across them

3. Alternative Analytical Approach

3.1 Effect Size Analysis (Cohen's d)

Every t-test verdict in the case is a binary decision: reject or fail to reject. What it does not tell you is **how big the effect is relative to the natural variability of the data.** Cohen's d provides that.

Hyp.	Effect (units)	Cohen's d	Magnitude	Faculty verdict	What effect size adds
H1	+2.2 min	**d ≈ 1.26**	**Very large**	Reject — fix process	Confirms urgency; this is not a marginal breach
H2	+0.007 pts	**d ≈ 0.04**	**Negligible**	Fail to reject	Confirms: this narrative is dead, not just inconclusive
H3	+₹18.9k/day	**d ≈ 3.54**	**Enormous**	Reject — investigate	Gap is not close to noise; cost study is justified
H4	+0.11 ppts	**d ≈ 0.46**	**Small-Medium**	Fail to reject	Gap is real enough to retest; budget freeze, not budget kill
H5	+₹21/week	**d ≈ 6.55**	**Exceptional**	Reject — roll out	This is one of the strongest effects in applied business stats
H6	−0.47 min	**d ≈ 0.51**	**Medium**	Reject (1-tail) — renegotiate	Statistically real, but operationally trivial vs H1's 2.2-min gap

Cohen's d = effect size / pooled standard deviation. Thresholds: d < 0.2 negligible, 0.2–0.5 small, 0.5–0.8 medium, > 0.8 large. H5's d ≈ 6.55 and H3's d ≈ 3.54 reflect pedagogically designed data and would be unusual in real-world applications.

3.2 Confidence Intervals as an Alternative Frame

Instead of asking "is this significant?", confidence intervals ask "what is the plausible range of the true effect?" For the three reject cases, the 95% CIs look like this:

H1: 95% CI for mean fulfillment time: [31.24 min, 33.16 min]. The entire interval is above 30 — not a single plausible value of the true mean is at or below the promise. This is stronger than just saying "reject H0."

H5: 95% CI for mean basket increase: [₹19.28, ₹22.85]. The loyalty program is adding somewhere between ₹19 and ₹23 per week per store. With 60 stores, that is roughly ₹69,000–82,000 per week in incremental revenue — which makes the rollout cost-benefit analysis very concrete.

H6: 95% CI for mean time reduction: [−0.96 min, +0.03 min]. Notice that the interval just touches zero on the upper end — which is exactly why this result would fail a two-tailed test. The software reduces delivery time, but we cannot rule out that the reduction is essentially zero.

3.3 What the Alternative Approaches Change

Sir's solution's binary verdicts are correct. The alternative framing does not change any of the six decisions. What it does change is the **confidence with which you make those decisions** and the quality of the conversation with the CEO. Telling the CEO that delivery times breach the promise by somewhere between 1.2 and 3.2 minutes (95% CI) is a more useful action brief than saying "t = 4.90, p < 0.05, reject H0."

4. Most Impactful Variable and Optimal Decision Set

4.1 Which Variable Has the Most Business Impact?

The direct answer: H5 (basket size / loyalty program) has the highest impact in terms of magnitude and certainty combined.

H5 shows a ₹21/week per-store increase across all 15 pilot stores with a t-statistic of 25.36 — effectively a certainty result. Extrapolated to 60 stores, the chain-wide weekly revenue gain is approximately **₹12,600 per week** or roughly ₹6.5 lakh per year from basket size alone, before accounting for frequency effects. No other variable in the dataset comes close in terms of a confirmed, positive, directly actionable effect.

That said, **H1 (delivery time) has the most strategic importance** even though the direction of its effect is negative — it is the variable most directly connected to Nandan Mart's core value proposition. A chain that cannot reliably deliver in 30 minutes cannot justify its delivery arm. H5 is the biggest win; H1 is the biggest problem.

4.2 Tail Choice Justification for Each Hypothesis

Hyp.	Business Claim	Tail	df	Cutoff	Justification
H1	Deliveries exceed 30 min	**Right (1-tail)**	14	$t > 1.761$	Being faster than 30 min needs no fix. Only a confirmed breach triggers operational change.
H2	CSAT has moved off 4.0	**Two-tailed**	14	$\|t\| > 2.145$	Both a rise (CX spend worked) and a fall (service problem) would change decisions. Either direction matters.
H3	Urban vs. suburban differ	**Two-tailed**	28	$\|t\| > 2.048$	If suburban outperforms urban, that changes strategy just as much as the reverse. Portfolio decision is direction-neutral.
H4	App converts better than web	**Right (1-tail)**	28	$t > 1.701$	Budget shifts to app only if app is confirmed better. If web is equal or better, the current split holds. Asymmetric decision.
H5	Loyalty pilot raised basket	**Right (1-tail)**	14	$t > 1.761$	Rollout is justified only by an increase. A decrease or no change means the pilot failed — same decision in both those cases: do not roll out.
H6	Route software cut times	**Left (1-tail)**	14	$t < -1.761$	Vendor's specific claim is a reduction. License renewal depends on that direction being confirmed. An increase would actually be a stronger case against renewal than a null result.

4.3 Recommended Decision Set

If Devika had to pick the minimum set of tests that cover the highest-stakes decisions:

Priority	Hypothesis	Result	Effect size	Decision
1	H5 — Loyalty rollout	**Reject H_0 (t = 25.36)**	d ≈ 6.55 (exceptional)	Roll out chain-wide immediately
2	H1 — Delivery breach	**Reject H_0 (t = 4.90)**	d ≈ 1.26 (large)	Fix dispatch process now
3	H6 — Route software	**Reject H_0 1-tail (t = −1.97)**	d ≈ 0.51 (medium; 28 sec)	Renegotiate license; do not renew at current price
4	H2 — CSAT narrative	**Fail to reject (t = 0.15)**	d ≈ 0.04 (negligible)	Kill the CX overhaul discussion
5	H4 — App vs. web	**Fail to reject (t = 1.27)**	d ≈ 0.46 (small-medium)	Freeze budget split; retest with n ≥ 50
6	H3 — Urban vs. suburban	**Reject H_0 (t = 9.59)**	d ≈ 3.54 (large)	Commission cost-to-serve study; do not close stores yet

5. Guardrails and Where t-Tests Break Down

The case study covers this section well, but it is worth restating the guardrails in my own words to confirm they are understood rather than just copied.

5.1 The Three Rules Before Running Any Test

Rule 1 — Identify the data structure first, not the test. The test family follows from the data. V7 and V8 are the same 15 stores measured twice — that is a paired test. V3 and V4 are different store-days — that is an independent test. Swapping these is not a small error; it produces a completely different (and wrong) test statistic.

Rule 2 — Fix the tail before looking at the numbers. H6 is the case study's own proof that these matters. The same dataset produces p = 0.034 one-tailed and p = 0.068 two-tailed. The first rejects, the second does not. If you look at the data first and then pick the tail that gives you significance, you have produced a meaningless result.

Rule 3 — Attach an effect size in business units to every verdict. A p-value is not an effect. A 28-second reduction and a ₹21/week increase are both "statistically significant" — but they are completely different business outcomes. The p-value says whether the effect is real; the effect size says whether the effect matters.

5.2 Where the Machinery Breaks Down

Small n and non-normal data: At n = 15, the t-test relies on the underlying data being approximately normal. All ten variables in this case look benign from the data dictionary, but a real deployment where, say, delivery times had a right skew from occasional extreme delays (60+ minutes) would call for Wilcoxon signed-rank (paired) or Mann-Whitney U (independent) as non-parametric alternatives.

Overlapping samples treated as independent: H3 compares urban and suburban stores. If some delivery zones overlap (an urban store and a nearby suburban store sharing the same customers), the independence assumption is violated, and the pooled t-test produces incorrect standard errors. The fix is either to control for the overlap in the study design or to use a mixed model.

Multiple testing inflation: Running six tests at $\alpha = 0.05$ each gives a roughly 26% chance of at least one false rejection if all six nulls were actually true ($1 - 0.95^6 \approx 0.265$). For this portfolio, where every test gates a pre-planned business decision, this is generally accepted. But if Devika's team had run 20 exploratory tests and presented only the five significant ones, a Bonferroni correction ($\alpha/6$ per test for six planned tests, so $\alpha \approx 0.0083$ each) or false discovery rate control would be necessary.

What the test cannot see: H3 cannot speak to profitability. H6 cannot speak to the software's routing benefits or fuel savings. H5 cannot say whether the basket increase is driven by actual consumption or just shifted timing. A t-test answers exactly the question encoded in its hypothesis — nothing adjacent to it.

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6. Summary Conclusion

The faculty solution is correct. All six t-statistics, standard errors, and p-values verified against the raw dataset. The framing table is the strongest part of the solution — the pre-commitment to tail structure before computing is what gives H6 its legitimacy and is the key habit this case is trying to build.

The most important variable is **H5 (basket size / loyalty program)** by a large margin in terms of confirmed positive effect. The most strategically urgent variable is **H1 (delivery time)** because it is the one variable where the company is currently failing its customers.

An alternative approach using effect sizes and confidence intervals does not change any of the six verdicts, but it makes the CEO conversation significantly more concrete — particularly for H5 (where the 95% CI translates directly into a weekly revenue range) and H6 (where the CI shows the improvement just barely rules out zero at the one-tailed level).

The minimum viable decision set is H1, H5, and H6 — act on the loyalty rollout, fix the delivery process, and renegotiate the route software license. The other three tests are valuable for closing down false narratives (H2), scoping follow-on studies (H3), and queuing a properly powered retest (H4).